

RESEARCH ARTICLE

Stability Analysis of Real-Time Hybrid Simulation with an Inerter-Type Experimental Substructure

Junjie Tao¹  | Oya Mercan² | Yuanfeng Duan³  | Guohua Xing¹

¹School of Civil Engineering, Chang'an University, Xi'an, China | ²Department of Civil and Mineral Engineering, University of Toronto, Toronto, Canada |

³College of Civil Engineering and Architecture, Zhejiang University, Hangzhou, China

Correspondence: Junjie Tao (junjie@chd.edu.cn)

Received: 26 June 2024 | **Revised:** 31 January 2025 | **Accepted:** 4 February 2025

Funding: This study is based upon work funded by the State Key Laboratory of Disaster Reduction in Civil Engineering (Grant No. SLDRCE24-04) and the National Natural Science Foundation of China (Grant No. 52478124, 52378318, and 52361165658).

Keywords: inerter | numerical damping | real-time hybrid simulation (RTHS) | spurious roots | stability | time delay

ABSTRACT

Although integrating inerters with conventional passive vibration control systems has shown enhanced performance in various studies, experimental investigations remain limited. Real-time hybrid simulation (RTHS) models the well-understood portion of a structure as the numerical substructure (NS) while physically testing the structural component of interest as the experimental substructure (ES). Testing the inerter as the ES in RTHS is considered cost-effective and less facility demanding. However, RTHS experiences time delay induced by actuator dynamics, risking instability if not managed effectively. Consequently, stability analysis is crucial for the successful implementation of RTHS. Previous studies primarily focused on RTHS stability, including a stiffness-type ES. The RTHS stability with an inerter-type ES, characterized by a large mass ratio relative to the NS, remains underexplored. To address this gap, this study analyzes the RTHS stability, including an inerter-type ES, implemented through various direct integration algorithms. Augmented state-space equations are employed to solve the roots of the discrete RTHS system considering different values of time delay. Virtual RTHSs are performed to validate the analytical investigation. The time delay is found to increase the order of the discrete RTHS system, yielding more spurious roots. Moreover, the time delay in RTHS with a stiffness-type ES primarily increases the magnitude of principal roots, whereas in RTHS with an inerter-type ES, it mainly amplifies the magnitude of spurious roots, potentially inducing instability. Both analytical and simulation results show that the spurious root-induced instability can be effectively mitigated by numerical damping.

1 | Introduction

Inerter is a two-terminal device capable of generating an inertial force that is proportional to the relative acceleration between its terminals. The associated proportionality constant is termed inertance, which can be interpreted as equivalent mass, measured in kilograms in the SI unit system. By utilizing specific mechanisms such as ball-screw [1], rack-and-pinion [2], and helical fluid [3], the inertance can be easily amplified. As a result, an inerter can produce a substantial inertial force without significantly increasing its physical mass. Compared to a conventional tuned mass

damper (TMD), an inerter is more space-efficient and does not introduce a large auxiliary mass. Moreover, the inerter is a two-terminal element, which enables it to be installed with other two-terminal springs and dashpots, forming various vibration control systems including inerter-based energy dissipators, inerter-based dynamic vibration absorbers, and inerter-based isolators. As the generated inertial force assists rather than resists motion, an inerter tends to provide an equivalent negative stiffness to the controlled structure, thereby reducing its natural frequency. Nevertheless, this reduction does not affect the structural static response since the inerter does not respond to static loading.

Because of the above attractive characteristics, inerters have been widely used to upgrade conventional vibration control systems for enhanced performance. Specifically, they are employed to increase the energy dissipation of energy dissipators. For instance, Ikago et al. [4] proposed a tuned viscous mass damper (TVMD), comprising a ball-screw inerter and a viscous damper, and validated its superior performance in comparison to a conventional viscous damper with the same damping coefficient in reducing the peak amplitude of the resonance curve for an earthquake-excited single-degree-of-freedom (SDOF) structure. This combination of the inerter and viscous damper was further utilized by Lu et al. [5] to mitigate the wind-induced vibrations of stay cables. Nakamura et al. [6] replaced the viscous damper with an electromagnetic damper, developing an electromagnetic inertial mass damper (EIMD), which allowed for controllable damping force through its terminal resistance. The EIMD demonstrated enhanced effectiveness in reducing story drifts and accelerations compared to conventional fluid viscous dampers and viscoelastic dampers. Li et al. [7] introduced the EIMD into damper-cable systems and found that the EIMD resulted in a first modal damping ratio almost 200% larger than the theoretical upper limit attainable with a viscous damper alone. In addition to being used with energy dissipators, inerters are employed alongside TMDs, constituting combined dynamic vibration absorbers, leveraging their efficiency in amplifying inertance to generate larger inertial forces. For example, Marian and Giaralis [8] proposed a configuration, namely the tuned mass-damper-inerter (TMDI), where one terminal of the inerter was connected to the TMD while the other terminal was linked to the host structure to suppress the oscillation of stochastically excited systems. It was analytically shown that the TMDI outperformed the classical TMD in minimizing displacement variances of an undamped SDOF structure subjected to white-noise excitations. Moreover, Giaralis and Taflanidis [9] confirmed this enhancement for multi-degree-of-freedom (MDOF) systems and concluded that the performance of TMDIs was more robust to structural uncertainties in comparison to TMDs. Sarkar and Fitzgerald [10] proposed a tuned mass damper fluid inerter (TMDFI), where a TMD was in series with a helical fluid inerter, to mitigate wind-induced vibrations in tall buildings. In a numerical simulation of a wind-excited 76-story building, the TMDFI was found to reduce the root-mean-square displacement of the roof by 65%, almost double the reduction achieved by a traditional TMD. Furthermore, inerters are also utilized to improve the seismic performance of base-isolated structures (BISs). In this regard, Zhao et al. [11] proposed a friction pendulum inerter system (FPIS). Wang et al. [12] incorporated the EIMD into the BISs below the isolation floor, while De Domenico and Ricciardi [13] introduced the TMDI as an alternative approach.

While numerous studies have shown the benefits of integrating inerters with traditional passive vibration control systems to mitigate structural vibrations and have proposed various design methods to optimize this integration in terms of inertance and system topology, the majority of these studies rely solely on numerical simulations. Moreover, as reviewed in ref. [14], existing experimental studies are predominantly small-scale tests focusing on inerter-based dissipators only. Large- and full-scale tests and experimental studies on inter-based dynamic absorbers and inter-based isolation systems remain limited, significantly impacting the real-world applications of these devices. One of the

main reasons for this lack of experimental study is the high cost and the demanding facility requirements involved.

Real-time hybrid simulation (RTHS) is a cyber-physical system initially proposed as an alternative to conventional shaking table testing for evaluating the seismic performance of structures during earthquakes [15–18]. In RTHS, the structural component of interest, characterized by complex hysteresis, is tested physically in real time, referred to as the experimental substructure (ES), while the remaining portion, with more well-studied properties, is modeled as the numerical substructure (NS). During testing, the responses of the NS are computed analytically using direct integration algorithms. In particular, the displacement boundary conditions at the interface between the ES and NS are imposed on the ES using actuators, ensuring deformation compatibility. The resulting force from the deformed ES is measured and fed back to the NS to maintain force equilibrium. This unique combination of experimental testing and numerical simulation makes RTHS cost- and space-efficient, effectively reducing the demand for experimental facilities to carry out an experimental study. Consequently, the application of RTHS for conducting experimental studies on inerter-based vibration control systems is a viable solution to address the challenges posed by high costs and facility shortages associated with conventional testing methods. Tao et al. [19] obtained the dynamic response of a base-isolated structure equipped with an inerter device through a series of RTHSs. In this RTHS configuration, the NS is a concentrated plasticity model of a nine-story building isolated with six lead rubber bearings, while the ES is a rack-and-pinion inerter (also known as the gyromass damper). The study experimentally validates the effectiveness of inerters in reducing maximum story drifts and base shears, while the optimization problems regarding inertance and topology remain unresolved.

To further advance the application of RTHS in experimental studies with inerters, it is crucial to thoroughly study the stability of RTHS with inerter-type ESs. Due to actuator dynamics [20], amplitude errors and time delay occur between the desired and measured deformation of the ES while the actuator imposes the desired boundary conditions, resulting in reduced experimental accuracy. Moreover, the time delay causes the measured force of the ES to lag behind the dynamic response of the NS, potentially affecting the stability of the RTHS system. Previous studies [21–24] have shown that the delayed restoring force from the ES introduces an equivalent negative damping to the RTHS system, potentially leading to instability. In contrast, the delayed inertial force introduces an equivalent positive damping ratio, stabilizing the RTHS system. This may imply that an RTHS with an inerter-type ES may not experience instability. However, as most previous studies focus on stiffness-type ESs, where the mass is negligible compared to their numerical counterparts, only the former conclusion is comprehensively validated. Few studies have examined the stability of RTHS while considering delayed inertial forces of significant magnitude. Maghareh et al. [22] represented an RTHS as a neutral or retarded delay differential equation (DDE). If the highest-order derivative in the DDE includes delay, which corresponds to an RTHS with delayed inertial forces, it is classified as a neutral DDE [25]. Otherwise, it is referred to as a retarded DDE. Huang et al. [26] investigated the distribution of characteristic roots for a neutral DDE and concluded that a neutral DDE has infinite pairs of conjugate

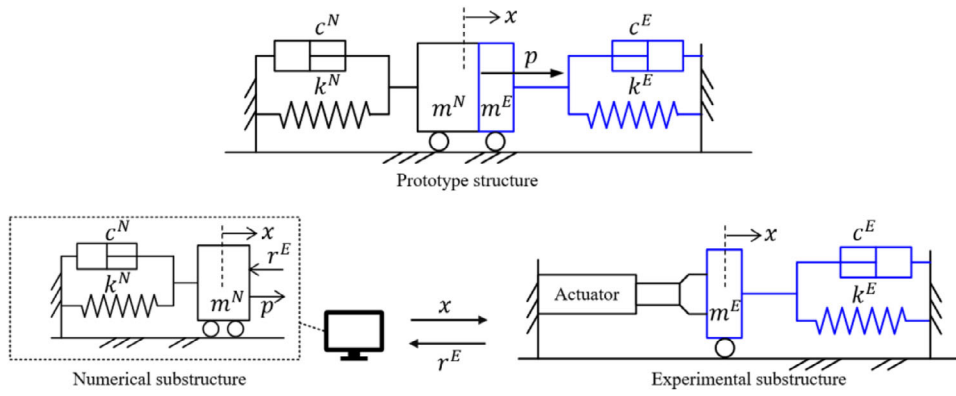


FIGURE 1 | Schematic of the SDOF RTHS.

roots, making them challenging to solve explicitly. Consequently, the authors employed the definite integral stability analysis method to determine the stability of the DDE without solving the characteristic equations. Despite these insights on neutral DDEs, refs. [22, 26] primarily focus on the equation of motion (EOM) of RTHS in the continuous-time domain rather than its implementation in the discrete-time domain, thereby not considering the effects of the integration algorithms utilized in RTHS. With the growing need to apply inerters as the ES in RTHS, where the inertance may equal or exceed the mass of the NS, a more practical investigation into the stability of the RTHS with an inerter-type ES is imperative.

This study analytically investigates the stability of an RTHS system with an inerter-type ES considering different values of time delay using augmented state-space equations. It is found that the time delay increases the order of the discrete RTHS system when implemented using direct integration algorithms, leading to additional spurious roots (i.e., eigenvalues). Although the increase in mass of the ES raises the damping ratio of the principal roots of the RTHS system, it also increases the magnitude of the spurious roots, thereby causing instability. The stability of the RTHS with an inerter-type ES using different direct integration algorithms is thoroughly investigated. Unlike the delayed restoring force, the instability induced by the delayed inertial force from an inerter-type ES is challenging to resolve by increasing the inherent structural damping ratio. However, it can be managed by using appropriate direct integration algorithms that include numerical damping. Virtual RTHSs are conducted to validate the conclusions obtained from the analytical investigation.

2 | Conventional Understanding of the Effects of Time Delay on RTHS Stability

In this section, the energy analysis of the RTHS system in ref. [21] is reviewed as an example of the conventional understanding of the effects of the time delay on RTHS stability. To simplify the demonstration, the analysis is presented using an SDOF structure. As illustrated in Figure 1, the prototype structure subjected to dynamic loading in an RTHS is partitioned into the NS and the ES, resulting in the following EOM:

$$\begin{aligned} m^N \ddot{x} + c^N \dot{x} + k^N x + m^E \ddot{x}^E + c^E \dot{x}^E + k^E x^E &= p \\ m &= m^N + m^E, c = c^N + c^E, k = k^N + k^E \end{aligned} \quad (1)$$

This can be further rewritten as

$$\begin{aligned} m^N \ddot{x} + c^N \dot{x} + k^N x &= p + r^E \\ r^E &= -m^E \ddot{x}^E - c^E \dot{x}^E - k^E x^E \end{aligned} \quad (2)$$

In the above equations, m , c , and k are the mass, viscous damping term, and stiffness, respectively, while x denotes the displacement responses. The superscripts N and E refer to the NS and ES, respectively. Additionally, p represents the external force, and r^E is the reaction force generated by the ES under the boundary condition $(x^E, \dot{x}^E, \ddot{x}^E)$, which is imposed by actuators.

Horiuchi et al. [21] analyzed the energy variation δE induced by r^E per period, for the RTHS system described in Equation (2), as follows:

$$\begin{aligned} \delta E &= \delta E^m + \delta E^c + \delta E^k \\ \delta E^m &= - \int_0^{2\pi/\omega} m^E \ddot{x}^E \dot{x} dt \\ \delta E^c &= - \int_0^{2\pi/\omega} c^E \dot{x}^E \dot{x} dt \\ \delta E^k &= - \int_0^{2\pi/\omega} k^E x^E \dot{x} dt \end{aligned} \quad (3)$$

where δE^m , δE^c , and δE^k represent the energy components associated with the mass, damping, and stiffness of the ES. To simplify the computation, the external loading is assumed to be harmonic, leading to RTHS steady-state responses as

$$\begin{aligned} x &= A \sin(\omega t) \\ x^E &= A \sin(\omega t - \omega \Delta t) \end{aligned} \quad (4)$$

where A is the maximum amplitude, ω is the circular frequency of the harmonic external loading, t is the time, and Δt is the time delay between x and x^E induced by actuator dynamics. Substituting Equation (4) into Equation (3) and assuming that Δt is small, the change in the total system energy per period, δE , can be computed as follows:

$$\begin{aligned} \delta E^m &= m^E A^2 \omega^3 \int_0^{2\pi/\omega} \sin(\omega t - \omega \Delta t) \cos(\omega t) dt = -m^E \Delta t A^2 \pi \omega^3 \\ \delta E^c &= -c^E A^2 \omega^2 \int_0^{2\pi/\omega} \cos(\omega t - \omega \Delta t) \cos(\omega t) dt = -c^E A^2 \pi \omega \\ \delta E^k &= -k^E A^2 \omega \int_0^{2\pi/\omega} \sin(\omega t - \omega \Delta t) \cos(\omega t) dt = k^E \Delta t A^2 \pi \omega \end{aligned} \quad (5)$$

The results in Equation (5) indicate that when Δt is positive (i.e., indicating a time delay), the restoring force (i.e., $k^E x^E$) of the ES introduces energy into the RTHS system, contributing negative damping with a magnitude of $-k^E \Delta t$. In contrast, the inertial force (i.e., $m^E \ddot{x}^E$) of the ES dissipates energy, acting as positive damping with a magnitude of $m^E \Delta t \omega^2$. If Δt is negative (i.e., indicating a time lead), the above effects will be reversed, though this scenario is rare in RTHS.

Therefore, in RTHS, a partitioning strategy with a small stiffness ratio of ES to NS and a large mass ratio of ES to NS is preferred, as it tends to stabilize the system with an overall positive damping ratio. Similar conclusions with respect to the effects of the stiffness and mass of ES on the RTHS stability also can be found in refs. [22, 23, 27]. According to the conclusions above, implementing an RTHS with an inerter-type ES characterized by a large inertance and a small stiffness may not encounter any instability, as the overall damping ratio remains positive based on Equation (5). However, the analysis in the following sections will yield different conclusions than those of the previous studies.

3 | Stability Analysis of SDOF RTHS Considering Different Direct Integration Algorithms

A limitation of previous RTHS stability analyses [21–24, 26, 27] is that they were conducted in the continuous-time domain, whereas RTHS is carried out in the discrete-time domain using direct integration algorithms. These analyses did not account for the effects of direct integration algorithms when evaluating the stability of RTHS under time delay. Moreover, in the continuous-time domain, time delay is expressed as $e^{-\tau s}$, where τ is the delay value and s is the Laplace variable. This exponential function significantly complicates the solution of characteristic functions and leads to an infinite pair of conjugate roots for neutral DDEs. In contrast, in the discrete domain, time delay is represented as z^{-k} , where k is the integer number of delayed time steps and z is the complex variable. The inclusion of z^{-k} increases the order of the difference equations and results in a finite number of characteristic roots that can be explicitly computed. Therefore, the stability analysis hereafter will focus on the discrete-time domain, calculating the characteristic roots of delay difference equations defined by various direct integration algorithms used in RTHS.

3.1 | Discrete EOM of SDOF RTHS

The discrete form of Equation (2) over a time interval of dt is expressed as

$$m^N \ddot{x}_{n+1} + c^N \dot{x}_{n+1} + k^N x_{n+1} + m^E \ddot{x}_{n+1}^E + c^E \dot{x}_{n+1}^E + k^E x_{n+1}^E = p_{n+1}, \quad (6)$$

where the subscript n indicates the time step number, yielding

$$x_{n+1} = x(ndt + dt), p_n = p(ndt + dt). \quad (7)$$

When solving Equation (6) using direct integration algorithms in RTHS, dt represents the integration time step. Although the reaction force generated by the ES is continuous during physical testing in RTHS, it is only available as discrete samples when fed

back to Equation (6). Therefore, it is reasonable to approximate that the delay Δt associated with the ES is a multiple of the integration time step dt in RTHS. This approximation gives

$$x_{n+1}^E = x_{n+1-k}, \dot{x}_{n+1}^E = \dot{x}_{n+1-k}, \ddot{x}_{n+1}^E = \ddot{x}_{n+1-k}, \quad (8)$$

where the notation k is an integer representing the number of time intervals by which the ES responses lag behind the NS responses, related by

$$kdt = \Delta t. \quad (9)$$

For instance, in the benchmark control problem for RTHS [20], the integration time step dt is predefined as 1/4096 s, while the time delay of the RTHS system without compensation is approximately 10.4 ms (i.e., $k \approx 43$) [28]. This delay is reduced to 4.9–5.1 ms (i.e., $k \approx 20$) when a proportional-integral tracking controller and a phase lead compensator are used [20], and it can be further reduced to less than 2 ms ($k < 9$) with more advanced compensation methods [28–31]. Consequently, the discrete EOM of an SDOF RTHS including time delay error is formulated as

$$m^N \ddot{x}_{n+1} + c^N \dot{x}_{n+1} + k^N x_{n+1} + m^E \ddot{x}_{n+1-k} + c^E \dot{x}_{n+1-k} + k^E x_{n+1-k} = p_{n+1}. \quad (10)$$

To solve Equation (10), direct integration algorithms are utilized to approximate the dynamic responses at consecutive time steps. As stability is an inherent property of the SDOF RTHS system and independent of input p_{n+1} , the EOM for free vibration is employed for stability analysis, written as

$$m^N \ddot{x}_{n+1} + c^N \dot{x}_{n+1} + k^N x_{n+1} + m^E \ddot{x}_{n+1-k} + c^E \dot{x}_{n+1-k} + k^E x_{n+1-k} = 0. \quad (11)$$

3.2 | Discrete State-Space Equations of SDOF RTHS Including Different Values of Time Delay

In this section, three direct integration algorithms widely used in RTHS, namely, the CR method [32], the KR- α method [33], and the Newmark- β method [34] are introduced to formulate the discrete state-space equations of RTHS considering different values of time delay. The CR method and the KR- α method are examples of unconditionally stable explicit methods, predicting the response at time step $n+1$ solely based on the information at time step n . In contrast, the Newmark- β method is a representative of unconditionally stable implicit methods, requiring information from both time steps n and $n+1$ to compute the response at time step $n+1$.

CR method:

The CR method formulates the prediction of displacement and velocity at time step $n+1$ as follows:

$$\begin{aligned} x_{n+1} &= x_n + dt \dot{x}_n + adt^2 \ddot{x}_n \\ \dot{x}_{n+1} &= \dot{x}_n + adt \ddot{x}_n \end{aligned} \quad (12)$$

To achieve unconditional stability, the parameter a is computed by

$$a = 4 / (4 + 4\zeta\omega_n dt + \omega_n^2 dt^2), \quad (13)$$

where ω_n and ζ are the structural natural frequency and the associated modal damping ratio. Substituting Equation (12) into Equation (11) yields

$$\begin{aligned} \hat{m}\ddot{x}_{n+1} + m^*\ddot{x}_n + c^*\dot{x}_n + k^*x_n + m^E\ddot{x}_{n+1-k} \\ + c^E\dot{x}_{n+1-k} + k^E x_{n+1-k} &= 0 \\ \hat{m} &= m^N \\ m^* &= ak^N dt^2 + ac^N dt \\ c^* &= c^N + k^N dt \\ k^* &= k^N. \end{aligned} \quad (14)$$

Based on Equations (12) and (14), the state-space equations of RTHS can be derived as

$$\mathbf{X}_{n+1} = \mathbf{A}\mathbf{X}_n. \quad (15)$$

Here, $\mathbf{X}$ is the state vector, $\mathbf{A}$ is the amplification matrix (i.e., the system matrix) characterizing the dynamics of the RTHS. The analytical solution of $\mathbf{X}$ at time step $n+1$ is computed as

$$\mathbf{X}_{n+1} = \mathbf{A}^n \mathbf{X}_0 \quad (16)$$

where $\mathbf{X}_0$ is the initial state vector. The solution is stable if and only if $\mathbf{A}^n$ is bounded. Using the Jordan normal form, it is equivalent to requiring that all the roots (i.e., the eigenvalues) of $\mathbf{A}$ must lie within the unit circle in the complex plane. For the RTHS using the CR method, the amplification matrix $\mathbf{A}$ for $k=1$ is formulated as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & dt & adt^2 \\ 0 & 1 & adt \\ -(k^* + k^E)/\hat{m} - (c^* + c^E)/\hat{m} - (m^* + m^E)/\hat{m} & & \end{bmatrix}. \quad (17)$$

For cases with larger values of time delay (i.e., $k > 1$), the formulae of $\mathbf{A}$ are provided in the appendix of this paper.

KR- α method:

The KR- α method is an unconditionally stable explicit method with controllable numerical damping, it rewrites the discrete EOM of RTHS as follows [35]:

$$m^N \widetilde{\ddot{x}}_{n+1} + c^N \widetilde{\dot{x}}_{n+1} + k^N \widetilde{x}_{n+1} + \widetilde{r}_{n+1}^E = 0, \quad (18)$$

where the sign $\widetilde{(\cdot)}$ represents the weighted variables:

$$\begin{aligned} \widetilde{\ddot{x}}_{n+1} &= (1 - \alpha_3) \ddot{x}_{n+1} + \alpha_3 \ddot{x}_n \\ \widetilde{\dot{x}}_{n+1} &= (1 - \alpha_f) \dot{x}_{n+1} + \alpha_f \dot{x}_n \\ \widetilde{x}_{n+1} &= (1 - \alpha_f) x_{n+1} + \alpha_f x_n \\ \widetilde{r}_{n+1}^E &= (1 - \alpha_f) r_{n+1}^E + \alpha_f r_n^E \\ r_{n+1}^E &= m^E \ddot{x}_{n+1} + c^E \dot{x}_{n+1} + k^E x_{n+1}. \end{aligned} \quad (19)$$

In ref. [35], $\widetilde{r}_{n+1}^E$ represents only the weighted restoring force from the ES, with the weighted inertial and damping forces of the ES lumped into the NS and computed numerically. While, to account for the effects of time delay with an inerter-type ES, the $\widetilde{r}_{n+1}^E$ is modified in Equation (19) to represent all the weighted inertial, damping, and restoring forces, assumed to be measured from the ES.

The KR- α method predicts the displacement and acceleration at time step $n+1$ using

$$\begin{aligned} x_{n+1} &= x_n + dt\dot{x}_n + \alpha_2 dt^2 \ddot{x}_n \\ \dot{x}_{n+1} &= \dot{x}_n + \alpha_1 dt \ddot{x}_n \end{aligned} \quad (20)$$

In the above equations, the parameters α_1 and α_2 define the variation of the velocity and displacement between time steps $n+1$ and n , while the parameters α_3 and α_f are introduced for additional numerical damping inspired by the generalized- α method [34]. These parameters are computed by

$$\begin{aligned} \alpha_1 &= 1 / (1 + 2\gamma_1 \zeta \Omega + \beta_1 \Omega^2), \alpha_2 = (0.5 + \gamma_1) \alpha_1 \\ \alpha_3 &= (\alpha_m + 2\alpha_f \gamma_1 \zeta \Omega + \alpha_f \beta_1 \Omega^2) / (1 + 2\gamma_1 \zeta \Omega + \beta_1 \Omega^2) \\ \gamma_1 &= 0.5 - \alpha_m + \alpha_f, \beta_1 = (1 - \alpha_m + \alpha_f)^2 / 4, \alpha_m \\ &= (2\rho_\infty - 1) / (\rho_\infty + 1), \alpha_f = \rho_\infty / (\rho_\infty + 1). \end{aligned} \quad (21)$$

here, ρ_∞ denotes the high-frequency spectral radius in the range of $[0, 1]$, Ω refers to the dimensionless circular frequency, defined as the product of ω_n and dt , termed dimensionless frequency. As the value of ρ_∞ decreases, greater numerical damping is introduced into the RTHS. Specifically, when $\rho_\infty = 1$, the KR- α method and the CR method are mathematically equivalent, with no numerical damping introduced. Based on Equations (18)–(20), the EOM of the RTHS for KR- α method including time delay is obtained as

$$m^N \widetilde{\ddot{x}}_{n+1} + c^N \widetilde{\dot{x}}_{n+1} + k^N \widetilde{x}_{n+1} + \widetilde{r}_{n+1-k}^E = 0, \quad (22)$$

leading to the associated amplification matrix $\mathbf{A}$ for $k=1$ as follows:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{6 \times 6} \\ \mathbf{A}_{11} &= \begin{bmatrix} 1 & dt & \alpha_2 dt^2 \\ 0 & 1 & \alpha_1 dt \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \\ a_{3,1} &= \frac{k^N - k^E (\alpha_f - 1)}{m^N (\alpha_3 - 1)} \\ a_{3,2} &= \frac{c^N - c^E (\alpha_f - 1) - k^N (\alpha_f - 1) dt}{m^N (\alpha_3 - 1)} \\ a_{3,3} &= \frac{m^N \alpha_3 - m^E (\alpha_f - 1) - k^N (\alpha_f - 1) \alpha_2 dt^2 - c^N (\alpha_f - 1) \alpha_1 dt}{m^N (\alpha_3 - 1)} \\ \mathbf{A}_{12} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{k^E \alpha_f}{m^N (\alpha_3 - 1)} & \frac{c^E \alpha_f}{m^N (\alpha_3 - 1)} & \frac{m^E \alpha_f}{m^N (\alpha_3 - 1)} \end{bmatrix} \end{aligned}$$

$$\mathbf{A}_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{A}_{22} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (23)$$

Similarly, for cases with larger values of time delay ($k > 1$), the formulae of $\mathbf{A}$ are provided in the appendix.

Newmark- β method:

The Newmark- β method defines the relationship between the displacement and velocity responses at time steps n and $n + 1$ as follows:

$$\begin{aligned} x_{n+1} &= x_n + dt\dot{x}_n + (0.5 - \beta)dt^2\ddot{x}_n + \beta dt^2\ddot{x}_{n+1} \\ \dot{x}_{n+1} &= \dot{x}_n + (1 - \gamma)dt\ddot{x}_n + \gamma dt\ddot{x}_{n+1}, \end{aligned} \quad (24)$$

where β and γ are two parameters determining stability and accuracy. The Newmark- β method is unconditionally stable and called the average acceleration method when $\beta = 0.25$ and $\gamma = 0.5$. Substituting Equation (24) into Equation (11), it yields

$$\begin{aligned} \hat{m}\ddot{x}_{n+1} + m^*\ddot{x}_n + c^*\dot{x}_n + k^*x_n + m^E x_{n-k+1} \\ + c^E \dot{x}_{n-k+1} + k^E \ddot{x}_{n-k+1} &= 0 \\ \hat{m} &= m^N + \gamma c^N dt + \beta k^N dt^2 \\ m^* &= (1 - \gamma)c^N dt + (0.5 - \beta)k^N dt^2 \\ c^* &= c^N + k^N dt \\ k^* &= k^N. \end{aligned} \quad (25)$$

Analogously, the amplification matrix $\mathbf{A}$ of the RTHS using the Newmark- β method for $k = 1$ is computed as follows:

$$\mathbf{A} = \begin{bmatrix} 1 - (\beta k^* dt^2) / \hat{m} & dt - (\beta c^* dt^2) / \hat{m} & (0.5 - \beta)dt^2 - (\beta m^* dt^2) / \hat{m} \\ -(\gamma k^* dt) / \hat{m} & 1 - (\gamma c^* dt) / \hat{m} & \gamma dt - \gamma m^* dt / \hat{m} \\ -(k^* + k^E) / \hat{m} & -(c^* + c^E) / \hat{m} & -(m^* + m^E) / \hat{m} \end{bmatrix}, \quad (26)$$

while the amplification matrices $\mathbf{A}$ for ($k > 1$) are given in the appendix.

Based on the observation of the amplification matrices $\mathbf{A}$ in Equations (17), (23), (26) and in appendix, it is apparent that as the introduced time delay increases, the dimension of the amplification matrix $\mathbf{A}$ expands significantly, leading to an increasing number of roots. The root types of $\mathbf{A}$ include zero roots, real roots, and complex conjugate pairs. Among these roots, the zero roots lead to the immediate elimination of states at a single time step, with no impact on the system stability. The real root typically results in exponential growth or decay of the states, depending on its position relative to the unit circle in the complex domain. The complex roots induce oscillatory behavior and can generally be expressed as

$$R_{1,2} = \sigma \pm i\varepsilon = \exp\left(\bar{\Omega}\left(-\bar{\xi} \pm i\sqrt{1 - \bar{\xi}^2}\right)\right) \quad (27)$$

where σ and ε denote the real and imaginary parts, respectively. Here, $\bar{\xi}$ represents the discrete equivalent damping ratio, and $\bar{\Omega} = \bar{\omega} dt$, where $\bar{\omega}_n$ is the discrete equivalent natural frequency. Specifically, the complex roots corresponding to the natural frequency of the SDOF system are considered principal. While,

the remaining complex roots, along with the real root, are considered spurious.

3.3 | Stability of RTHS with an Inerter-Type ES

To ensure the stability of the discrete RTHS system, all the roots of its amplification matrix $\mathbf{A}$ must lie within the unit circle in the complex plane. Since the spectral radius of a square matrix is defined as the maximum magnitude of its roots, this stability criterion requires that the spectral radius of $\mathbf{A}$ does not exceed one. In this study, the roots of $\mathbf{A}$ are calculated via eigenvalue analysis using MATLAB based on Equations (17), (23), (26) and the equations in appendix. Three parameters are defined as follows:

$$\lambda_m = m^E / m, \lambda_k = k^E / k, \lambda_c = c^E / c \quad (28)$$

where λ_m , λ_k , and λ_c represent the proportions of the mass, stiffness, and damping of the ES relative to the prototype structure, respectively. Notably, in the RTHS with an inerter-type ES, the value of λ_m is significantly greater than those of λ_k and λ_c .

In Figures 2–4, the magnitudes of the non-zero roots of $\mathbf{A}$ for the SDOF RTHS systems with an inerter-type ES, using the CR method, the KR- α method, and the Newmark- β method, are plotted against the dimensionless circular frequency Ω ($\Omega = \omega_n dt$), considering different values of time delay. The ρ_∞ in the KR- α method is defined as 0.75 to include numerical damping, while the β and γ in the Newmark- β method are set to 0.25 and 0.5, respectively, converting it into the average acceleration method. An inherent structural damping ratio of 2% is defined, and the inerter-type ES is assumed to have the properties: $\lambda_m = 0.3$, $\lambda_k = 0$, and $\lambda_c = 0$, providing an inertial force only. In the figures, the solid red lines denote the spectral radii for the RTHS system without any time delay, indicating an ordinary dynamic system solved by the direct integration algorithms. Whereas the solid black lines and dashed grey lines represent the magnitudes of the principal roots and the spurious roots for the RTHS systems including time delay, respectively. Their maximum values are plotted as spectral radii with dot yellow lines.

The results in Figures 2–4 indicate that the CR method, the KR- α method, and the Newmark- β method are all unconditionally stable when applied to solve an ordinary dynamic system without any time delay introduced. However, as the time delay increases from 0 to $2dt$, $4dt$, and $8dt$, the number of spurious roots in the dynamic systems increases accordingly. Although the delayed inertial force appears to stabilize the system by decreasing the magnitudes of the principal roots as Ω increases, it results in peaks in the magnitude of the spurious roots, exceeding the stability limit. Consequently, the RTHS system becomes conditionally stable when the time delay is introduced.

To quantify this conditional stability, the spurious roots whose magnitudes first exceed the stability limit are denoted by ρ_c , as shown in Figures 2–4. Moreover, the value of Ω at the crossing point is represented as Ω_c . Consequently, the range $[0, \Omega_c]$ represents the stability region. It is evident that as the introduced time delay increases, Ω_c decreases, approaching zero, thereby reducing the system stability region.

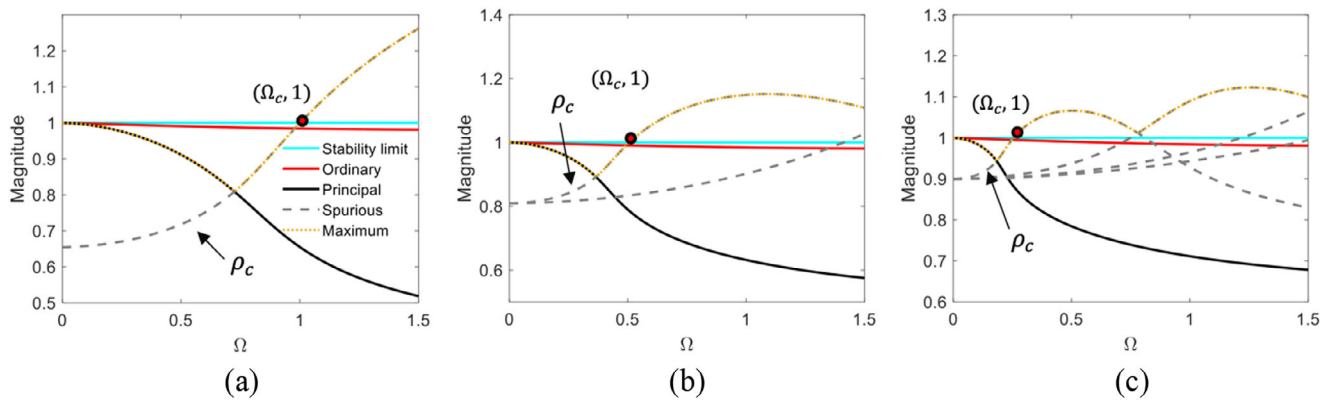


FIGURE 2 | Magnitudes of the roots for the SDOF RTHS systems ($\lambda_m = 0.3$, $\lambda_k = 0$, $\lambda_c = 0$) using the CR method considering different values of time delay (a) $\Delta t = 2dt$, (b) $\Delta t = 4dt$, (c) $\Delta t = 8dt$.

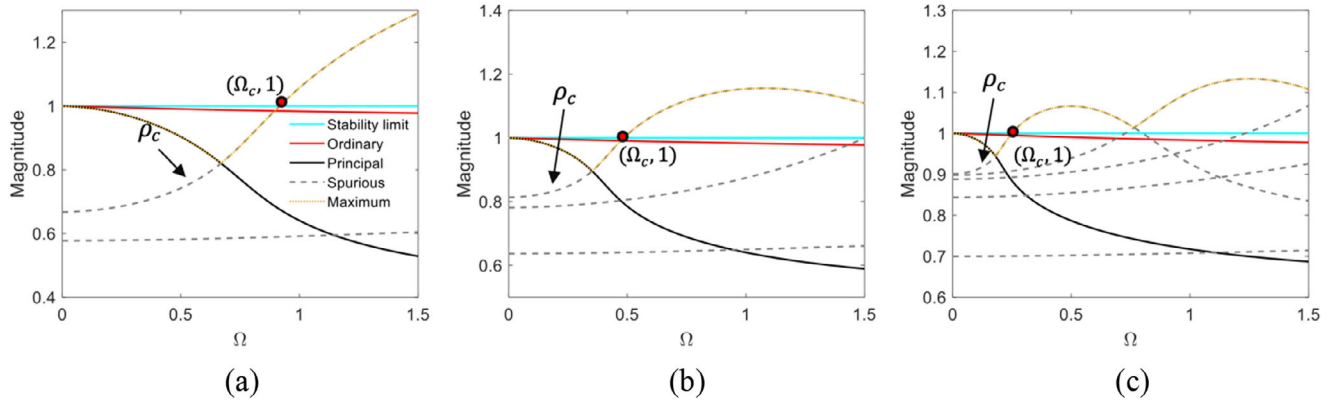


FIGURE 3 | Magnitudes of the roots for the SDOF RTHS systems ($\lambda_m = 0.3$, $\lambda_k = 0$, $\lambda_c = 0$) using the KR- α method considering different values of time delay (a) $\Delta t = 2dt$, (b) $\Delta t = 4dt$, (c) $\Delta t = 8dt$.

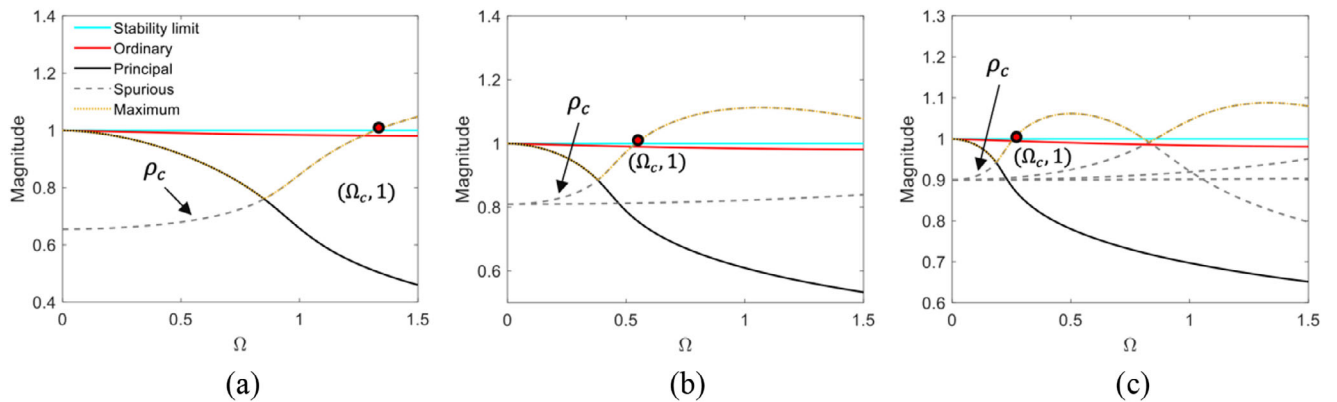


FIGURE 4 | Magnitudes of the roots for the SDOF RTHS systems ($\lambda_m = 0.3$, $\lambda_k = 0$, $\lambda_c = 0$) using the Newmark- β method considering different values of time delay (a) $\Delta t = 2dt$, (b) $\Delta t = 4dt$, (c) $\Delta t = 8dt$.

Figures 5 and 6 present the magnitudes and discrete equivalent damping ratios of the principal roots for the RTHS systems including an inerter-type ES with various values of λ_m , for the CR method, the KR- α method, and the Newmark- β method, respectively. While, Figure 7 illustrates the magnitude of the spurious roots ρ_c to analyze the effects of λ_m on the RTHS stability region. The time delay Δt is considered dt here to highlight the

influence of the ES mass on the characteristics of the RTHS systems.

As depicted in Figures 5 and 6, an increase in the ES mass decreases the magnitudes and increases the damping ratios of the principal roots. This observation is consistent with the conventional understanding of how ES mass affects RTHS stability, as

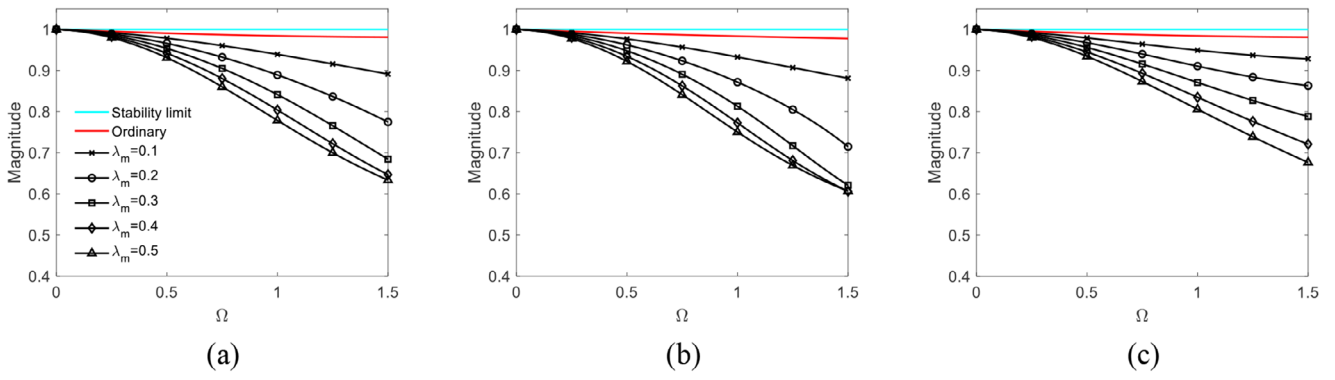


FIGURE 5 | Magnitudes of the principal roots for the SDOF RTHS systems ($\Delta t = dt$) including an inerter-type ES with variable λ_m (a) the CR method, (b) the KR- α method, (c) the Newmark- β method.

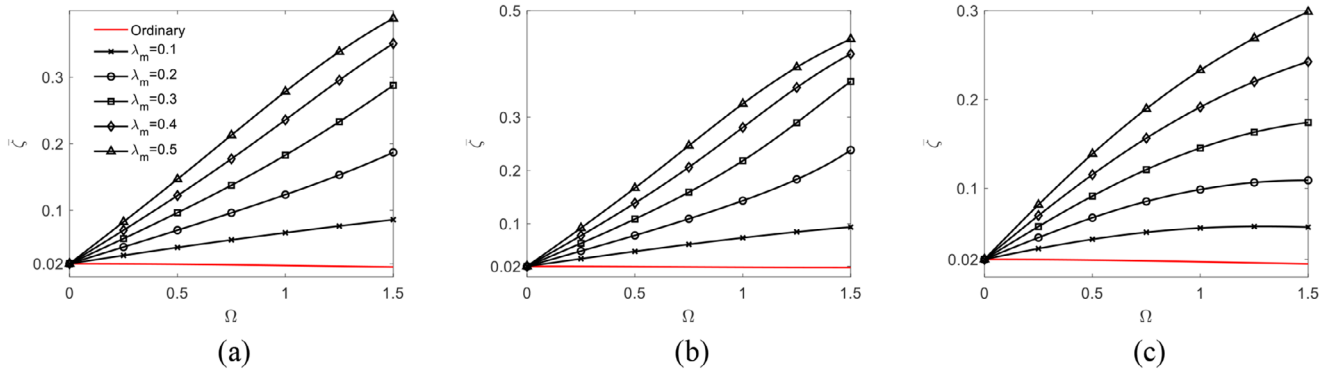


FIGURE 6 | Discrete equivalent damping ratios of the principal roots for the SDOF RTHS systems ($\Delta t = dt$) including an inerter-type ES with variable λ_m (a) the CR method, (b) the KR- α method, (c) the Newmark- β method.

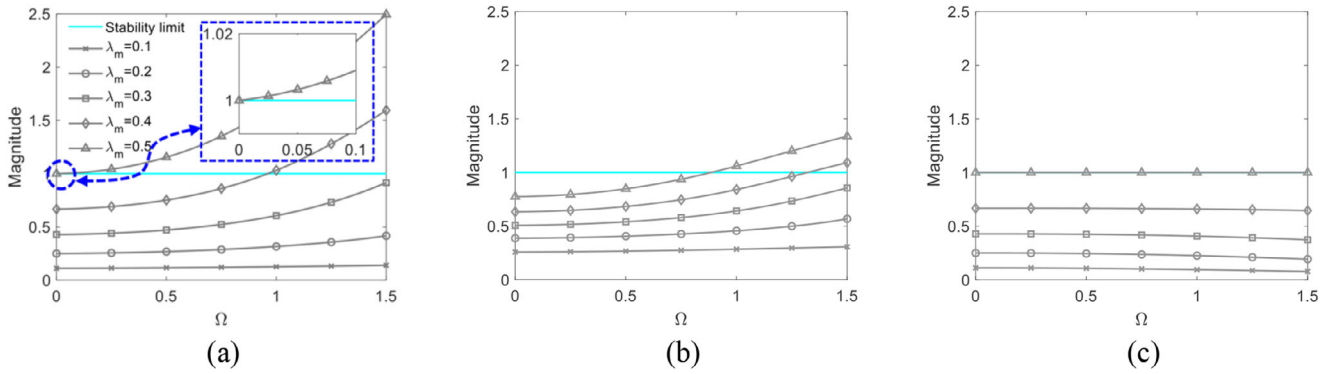


FIGURE 7 | Magnitudes of spurious roots ρ_c for the SDOF RTHS systems ($\Delta t = dt$) including an inerter-type ES with variable λ_m (a) the CR method, (b) the KR- α method, (c) the Newmark- β method.

demonstrated by the previous studies. However, Figure 7 reveals that increasing the ES mass leads to the opposite effect on the magnitude of the spurious roots. As λ_m increases from 0.1 to 0.5, the magnitudes of the spurious roots consistently increase, significantly reducing the system stability region $[0, \Omega_c]$. Specifically, when $\lambda_m = 0.5$, the magnitude of the spurious roots for the RTHS system using the CR method are greater than one, regardless of the value of Ω . Consequently, the RTHS implemented with the CR method becomes unstable irrespective of the value of Ω . For the Newmark- β method ($\beta = 0.25$ and

$\gamma = 0.5$), when $\lambda_m = 0.5$, the RTHS system is marginally stable, with the magnitude of the spurious roots ρ_c exactly equal to one. Any further increase in λ_m may lead to instability. Given the defined relationship $\Delta t = dt$, the above stability issues with the CR method and the Newmark- β method are challenging to mitigate merely by reducing the time delay using a compensation method. To ensure stability, complete elimination of the time delay is required (i.e., $\Delta t = 0$), which is challenging considering experimental uncertainties and disturbances. Moreover, as time delay increases, such as the cases considered in Figures 2–4,

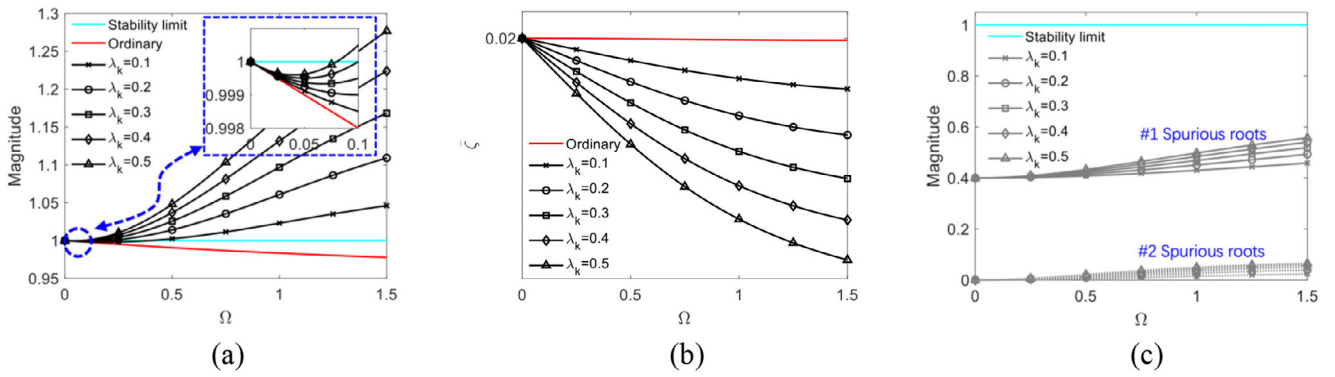


FIGURE 8 | Characteristics of the SDOF RTHS system ($\Delta t = dt$) using the KR- α method including a stiffness-type ES with variable λ_k , (a) magnitudes of the principal roots, (b) discrete equivalent damping ratios of the principal roots, (c) magnitudes of the spurious roots.

where $\Delta t = 2dt$, $\Delta t = 4dt$, $\Delta t = 8dt$, the aforementioned challenges worsen. In contrast, the KR- α method yields the spurious roots ρ_c with magnitude less than one when $\lambda_m = 0.5$, providing a nonzero stability region $[0, \Omega_c]$. The introduced numerical damping proves effective in mitigating the consistent increase in the magnitude of the spurious roots ρ_c with respect to Ω , allowing for conditional stability if dt is properly defined. In conclusion, due to the introduction of numerical damping, the KR- α method is more suitable for RTHS with an inerter-type ES than the CR method and the Newmark- β method, particularly when λ_m is large.

To highlight the differences between the effects of the ES mass and the ES stiffness on RTHS stability in response to time delay, the magnitudes and damping ratios of the roots for the RTHS systems with a stiffness-type ES ($\lambda_m = 0$, $\lambda_c = 0$), using the KR- α method are presented in Figure 8. As observed from the figure, when time delay is introduced in the RTHS, the ES stiffness primarily affects the principal roots of the system, with negligible effects on the spurious roots. Increasing λ_k leads to an increase in the magnitudes and a decrease in the damping ratios of the principal roots, which is opposite to the effects of λ_m . The instability of the RTHS with a stiffness-type ES is determined by the magnitudes of its principal roots, whereas the instability of the RTHS with an inerter-type ES is determined by the magnitude of its spurious root ρ_c . Despite the increase in the magnitudes of principal roots, a non-zero stability region remains, albeit reduced, as depicted in Figure 8a. This indicates that the principal root-induced instability is conditional and can be resolved by using a smaller time step size dt and reducing time delay through compensation methods. However, the increase in the magnitudes of the spurious roots is consistent and may lead to the elimination of the stability region, as depicted in Figure 7a. As a result, the spurious root-induced instability is determined regardless of Ω .

Figure 9 demonstrates the effects of inherent structural damping and numerical damping on the magnitudes of the spurious roots ρ_c for the RTHS including an inerter-type ES, where the mass ratio λ_m is considered 0.5 and the time delay Δt is defined as $2dt$. The KR- α method is considered here for its superiority in introducing numerical damping. As illustrated in the figure, the magnitudes of the spurious roots ρ_c are not sensitive to the variation of the structural inherent damping. Moreover, as the inherent damping ratio ζ increases from 2% to 5%, the increase in

magnitudes is accelerated, as indicated in Figure 9b. In contrast, the numerical damping noticeably affects the magnitudes of the spurious roots ρ_c . As the numerical damping increases (i.e., ρ_∞ decreases), ρ_c effectively consistently decreases, extending the stability region $[0, \Omega_c]$.

Based on the results presented in Section 3.3, the following observations regarding the stability of an SDOF RTHS including an inerter-type ES can be noted:

1. When an RTHS system is implemented in the discrete-time domain using direct integration algorithms, time delay increases the order of the discrete system, resulting in an increasing number of spurious roots.
2. The mass of the ES responds to time delay by decreasing the magnitudes of principal roots, increasing the principal damping ratio, and increasing the magnitudes of spurious roots. The increase in magnitudes of spurious roots contributes to potential instability.
3. When $\lambda_m > 0.5$, the RTHS is unstable regardless of the value of Ω , if implemented using direct integration algorithms that include zero numerical damping. Increasing inherent structural damping and reducing integration time step size are ineffective in resolving this instability.

4 | Extension to MDOF RTHS Using Different Direct Integration Algorithms

Due to the wide variety of partitioning cases, it is challenging to derive a mathematical representation of an MDOF RTHS for various integration algorithms without specific assumptions. To generalize the previous stability analysis and provide insights into the effects of time delay in an MDOF RTHS with an inerter-type ES, this study assumes that the partitioning cases follow the relationship:

$$\mathbf{M}^E = \lambda_m \mathbf{M}, \mathbf{K}^E = \lambda_k \mathbf{K}, \mathbf{C}^E = \lambda_c \mathbf{C}, \quad (29)$$

as illustrated in Figure 10. In the above equation, $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the mass, classic damping, and stiffness matrices of the prototype MDOF structure, respectively. The scalars λ_m , λ_k , and λ_c denote

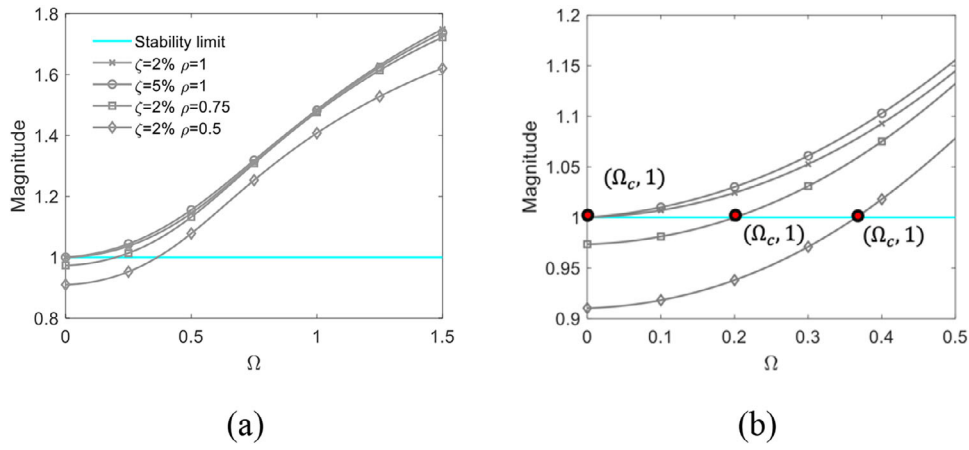


FIGURE 9 | Effects of inherent structural damping and numerical damping on the SDOF RTHS system using the KR- α method including an inerter-type ES ($\lambda_m = 0.5$, $\Delta t = 2dt$), (a) magnitudes of spurious roots ρ_c , (b) close-up view.

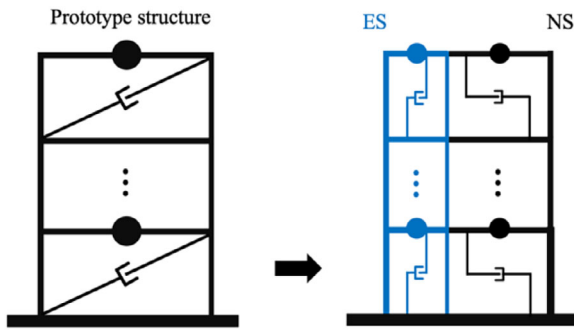


FIGURE 10 | Partitioning case considered for the MDOF RTHS system.

the proportions of the mass, damping, and stiffness matrices of the ES relative to those of the prototype structure.

Consequently, the stability analysis for an SDOF RTHS, described in Section 3, can be extended to an MDOF RTHS using the modal superposition. The discrete EOM for free vibration of an MDOF RTHS can be represented by

$$\begin{aligned} \underline{M}^N \ddot{\mathbf{X}}_{n+1} + \underline{C}^N \dot{\mathbf{X}}_{n+1} + \underline{K}^N \mathbf{X}_{n+1} \\ + \underline{M}^E \ddot{\mathbf{X}}_{n+1-k}^E + \underline{C}^E \dot{\mathbf{X}}_{n+1-k}^E + \underline{K}^E \mathbf{X}_{n+1-k}^E = 0, \end{aligned} \quad (30)$$

where $\mathbf{X}$ is the displacement response vector. Assuming that:

$$\mathbf{X}_{n+1} = \Phi \mathbf{Y}_{n+1}, \quad (31)$$

Equation (30) can be revised as follows:

$$\begin{aligned} \underline{M}^N \ddot{\mathbf{Y}}_{n+1} + \underline{C}^N \dot{\mathbf{Y}}_{n+1} + \underline{K}^N \mathbf{Y}_{n+1} + \underline{M}^E \ddot{\mathbf{Y}}_{n+1-k}^E \\ + \underline{C}^E \dot{\mathbf{Y}}_{n+1-k}^E + \underline{K}^E \mathbf{Y}_{n+1-k}^E = 0, \end{aligned} \quad (32)$$

where Φ is the eigenvector matrix of the prototype MDOF structure defined by $\underline{M}$ and $\underline{K}$, $\mathbf{Y}$ is the modal coordinate vector, and $\underline{M}$, $\underline{C}$, and $\underline{K}$ are the diagonal mass, damping, and stiffness matrices, obtained by premultiplying and postmultiplying the original mass, damping, and stiffness matrices represented in

physical coordinates by Φ^T and Φ . The EOM for each mode of free vibration is then expressed as

$$\begin{aligned} \underline{m}_{ii}^N \ddot{y}_{i,n+1} + \underline{c}_{ii}^N \dot{y}_{i,n+1} + \underline{k}_{ii}^N y_{i,n+1} \\ + \underline{m}_{ii}^E \ddot{y}_{i,n+1-k}^E + \underline{c}_{ii}^E \dot{y}_{i,n+1-k}^E + \underline{k}_{ii}^E y_{i,n+1-k}^E = 0, \end{aligned} \quad (33)$$

where $\underline{m}_{ii}$, $\underline{c}_{ii}$, and $\underline{k}_{ii}$ are the i -th diagonal terms of the diagonalized $\underline{M}$, $\underline{C}$, and $\underline{K}$, and $y_{i,n+1}$ is the modal coordinate of i th mode at time step $n+1$.

Accordingly, the stability analysis of an MDOF RTHS system using different direct integration algorithms can be evaluated for each vibration mode through the modal amplification matrix $\mathbf{A}_i$. Analogous to the computation of $\mathbf{A}$ in Section 3, the modal amplification matrices $\mathbf{A}_i$ for the CR method, the KR- α method, and the Newmark- β method can be derived from Equations (33), (12), (18–20), and (24). The formulae of $\mathbf{A}_i$ are nearly identical to those of $\mathbf{A}$, with all model-based coefficients converted to their modal forms [32, 33]. For conciseness, the formulae of $\mathbf{A}_i$ including a single time step delay (i.e., $k = 1$) are presented here for the CR method, the KR- α method, and the Newmark- β method:

CR method

$$\begin{aligned} \mathbf{A}_i = \begin{bmatrix} 1 & dt & \underline{a}dt^2 \\ 0 & 1 & \underline{a}dt \\ -\left(k^* + \underline{k}_{ii}^E\right)/\hat{m} - \left(c^* + \underline{c}_{ii}^E\right)/\hat{m} - \left(m^* + \underline{m}_{ii}^E\right)/\hat{m} & & \end{bmatrix} \\ \hat{m} = \underline{m}_{ii}^N \\ m^* = \underline{a} \underline{k}_{ii}^N dt^2 + \underline{a} \underline{c}_{ii}^N dt \\ c^* = \underline{c}_{ii}^N + \underline{k}_{ii}^N dt \\ k^* = \underline{k}_{ii}^N \end{aligned} \quad (34)$$

KR- α method

$$\begin{aligned}
 \mathbf{A}_i &= \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{6 \times 6} \\
 \mathbf{A}_{11} &= \begin{bmatrix} 1 & dt & \underline{\alpha}_2 dt^2 \\ 0 & 1 & \underline{\alpha}_1 dt \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \\
 a_{3,1} &= \frac{k_{ii}^N - k_{ii}^E (\alpha_f - 1)}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} \\
 a_{3,2} &= \frac{c_{ii}^N - c_{ii}^E (\alpha_f - 1) - k_{ii}^N (\alpha_f - 1) dt}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} \\
 a_{3,3} &= \frac{\underline{m}_{ii}^N \underline{\alpha}_3 - \underline{m}_{ii}^E (\alpha_f - 1) - k_{ii}^N (\alpha_f - 1) \underline{\alpha}_2 dt^2 - c_{ii}^N (\alpha_f - 1) \underline{\alpha}_1 dt}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} \\
 \mathbf{A}_{12} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{k_{ii}^E \alpha_f}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} & \frac{c_{ii}^E \alpha_f}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} & \frac{\underline{m}_{ii}^E \alpha_f}{\underline{m}_{ii}^N (\underline{\alpha}_3 - 1)} \end{bmatrix} \\
 \mathbf{A}_{21} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{A}_{22} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned} \quad (35)$$

Newmark- β method

$$\begin{aligned}
 \mathbf{A}_i &= \begin{bmatrix} 1 - (\beta k^* dt^2) / \hat{m} & dt - (\beta c^* dt^2) / \hat{m} & (0.5 - \beta) dt^2 - (\beta m^* dt^2) / \hat{m} \\ -(\gamma k^* dt) / \hat{m} & 1 - (\gamma c^* dt) / \hat{m} & \gamma dt - \gamma m^* dt / \hat{m} \\ -(k^* + k_{ii}^E) / \hat{m} & -(c^* + c_{ii}^E) / \hat{m} & -(m^* + \underline{m}_{ii}^E) / \hat{m} \end{bmatrix} \\
 \hat{m} &= \underline{m}_{ii}^N + \gamma c_{ii}^N dt + \beta k_{ii}^N dt^2 \\
 m^* &= (1 - \gamma) c_{ii}^N dt + (0.5 - \beta) k_{ii}^N dt^2 \\
 c^* &= c_{ii}^N + k_{ii}^N dt \\
 k^* &= k_{ii}^N
 \end{aligned} \quad (36)$$

Here, the model-based integration coefficients $\underline{a}$ for the CR method and α_i ($i = 1, 2, 3$) for the KR- α method are computed using Equations (13) and (21) with modal frequency ω_i and damping ratio ζ_i . Although the cases for $k > 1$ are not presented here, the similarity between the representation of $\mathbf{A}_i$ and $\mathbf{A}$ for $k = 1$ can be readily extended to the cases where $k > 1$. Consequently, the conclusions for the stability analysis in Section 3.3 also apply to the MDOF RTHS system for each vibration mode.

5 | Numerical Validation

While experiments provide the most realistic validation, they are not ideal for examining stability due to the potential risk of damaging servo-hydraulic testing equipment and specimens.

Therefore, three numerical examples are presented here to validate the findings from the stability analysis presented in the earlier sections.

5.1 | Example 1

In the first example, a portal frame, presented in Figure 11a and characterized by a fundamental period of 0.7 s and a damping ratio of 2 %, was modeled as an SDOF NS. An inerter installed between the floor and ground through braces was defined as the ES. The ES was assumed to provide only an experimental inertial force, given by

$$r^E = M^I \ddot{x}^E, \quad (37)$$

where M^I is the equivalent mass (i.e., the inertance) of the inerter and $\ddot{x}^E$ is the measured acceleration. The relationship between the equivalent mass of the ES and the mass of the NS was defined as

$$m^I = \lambda_m m, \quad m^N = (1 - \lambda_m) m, \quad mg = W = 1000 \text{ kN}. \quad (38)$$

To avoid potential numerical instability induced by simulating actuator dynamics using a high-order transfer function model and to focus on the effects of the time delay on the discrete RTHS system stability, the actuator dynamics in RTHS are simplified as a pure time delay, which gives

$$\begin{aligned}
 \ddot{x}_{n+1}^E &= \ddot{x}_{n+1-k} \\
 r_{n+1}^E &= m^I \ddot{x}_{n+1-k}^E.
 \end{aligned} \quad (39)$$

In these simulations, the CR method, the KR- α method, and the Newmark- β method were implemented with a time step of 1 ms. Assuming that a satisfactory, albeit imperfect, compensation was applied, the actuator-induced time delay was reduced to a single time step (i.e., $\Delta t = dt$). Consequently, the measured inertial force experienced a delay of 1 ms. Figure 11b presents the Simulink diagram of the performed simulations, where the actuator dynamics were approximated by a Z transform operator.

To enhance understanding and ensure the repeatability of the conducted simulations, the implementations of the CR method, KR- α method, and Newmark- β method for Example 1 are detailed in Figure 12. As shown, the simulation in each time loop includes three steps: (1) computing the delayed inertial force using $r_{n+1}^E = m^I \ddot{x}_n$, (2) calculating x_{n+1} and $\dot{x}_{n+1}$ based on the prediction schemes predefined in direct integration algorithms, and (3) determining $\ddot{x}_{n+1}$ by solving the EOM. In the CR method and the KR- α method, the delayed inertial force r_{n+1}^E is required only for Step 3. However, in the Newmark- β method, r_{n+1}^E is needed for both Steps 2 and 3. For linear structural systems, the implementation of the Newmark- β method is modified [36] to permit solutions without iterations.

Table 1 presents the simulation matrices for Example 1 using various direct integration algorithms. The JR Takatori record from the Kobe 1995 earthquake is employed as the dynamic input. Using the methods demonstrated in Section 3, the stability of each

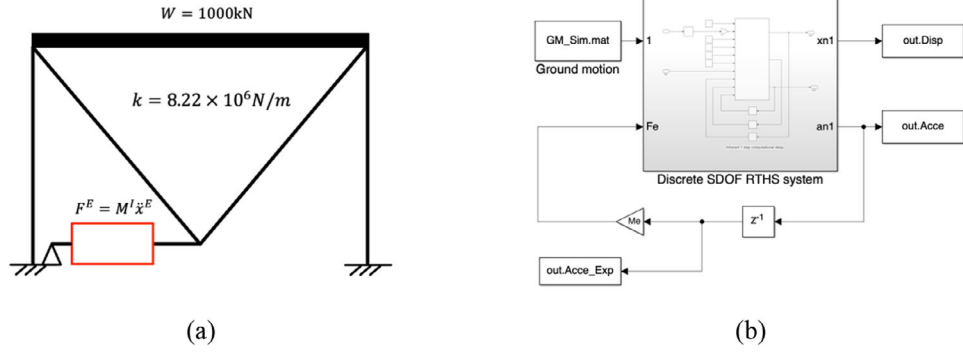


FIGURE 11 | Schematic of Example 1 (a) Prototype structure (b) Simulink model.

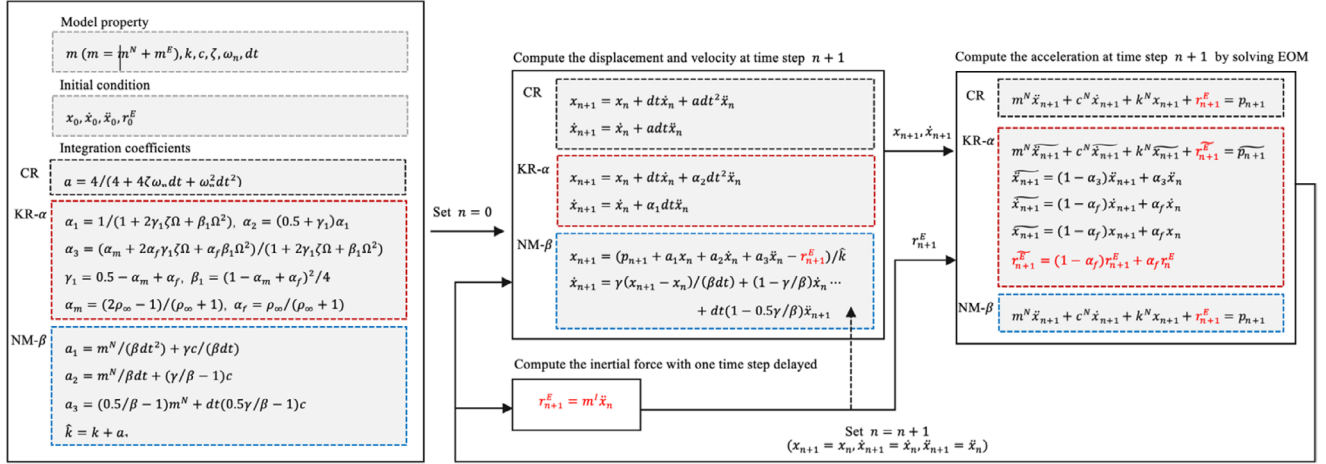


FIGURE 12 | Detailed implementations of the various integration algorithms in the vRTHS.

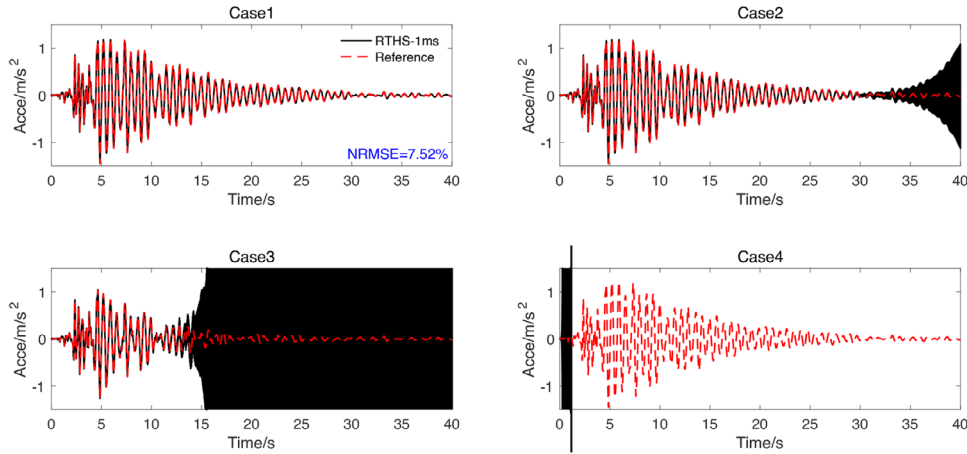


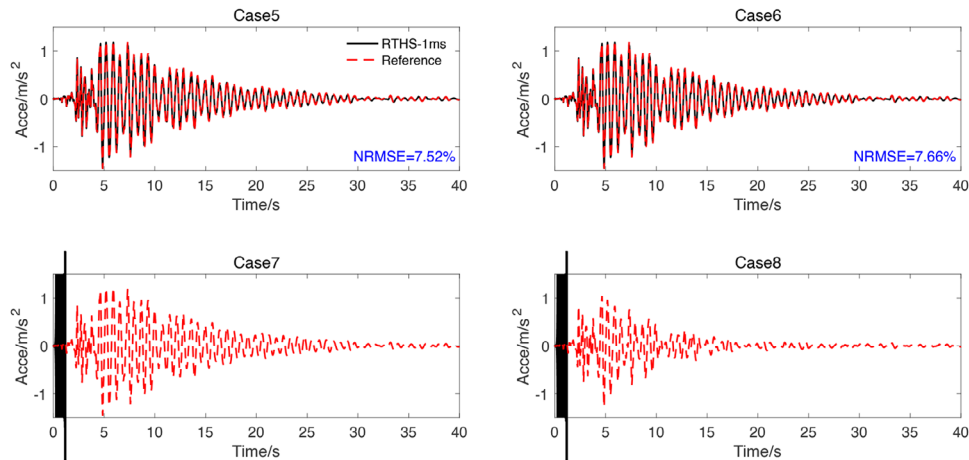
FIGURE 13 | Time histories of the acceleration in the simulations using the CR method.

simulation is predicted and listed in Table 1. The acceleration time histories of all the simulations are illustrated in Figures 13–15. Specifically, according to the previous stability analysis, $\lambda_m = 0.5$ is identified as the critical mass ratio to ensure stability for the RTHSs implemented with the CR method and the Newmark- β method. To validate this limit, the simulations using the CR method and the Newmark- β method were performed with λ_m values of 0.49, 0.5, and 0.51. On the other hand, the simulations

using the KR- α method were carried out with λ_m values set to 0.5, 0.55, and 0.6 to demonstrate the effectiveness of the numerical damping in mitigating the instability induced by spurious roots. Moreover, the results obtained through ordinary linear time history analyses (i.e., no time delay) are provided in the figures as the reference. The normalized root mean square errors (NRMSEs) [21] between the vRTHS and reference results are also labeled in the figures.

TABLE 1 | Simulation matrices of Example 1.

Case #	Direct integration algorithms	λ_m	ζ	ρ_∞	Prediction
1	CR method	0.49	2%	—	Stable
2		0.5	2%	—	Unstable
3		0.5	5%	—	Unstable
4	Newmark- β method	0.51	2%	—	Unstable
5		0.49	2%	—	Stable
6		0.5	2%	—	Stable
7		0.51	2%	—	Unstable
8	KR- α method	0.51	5%	—	Unstable
9		0.5	2%	0.85	Stable
10		0.5	2%	0.75	Stable
11		0.55	2%	0.85	Stable
12		0.55	2%	0.75	Stable
13		0.6	2%	0.85	Unstable
14		0.6	2%	0.75	Stable

**FIGURE 14** | Time histories of the acceleration in the simulations using the Newmark- β method.

The time histories provided in Figures 13–15 fully agree with the stability prediction listed in Table 1. Due to the lack of numerical damping, the CR method and the Newmark- β method are unable to ensure a stable RTHS when the mass ratio of the inerter-type ES exceeds 0.5. In particular, the time histories of Cases 2–3 and Cases 7–8 confirm that increasing inherent structural damping is ineffective in resolving the instability induced by the spurious roots. Moreover, the comparison between the results in Case 2 and Case 3 indicates that increasing the inherent structural damping deteriorates the stability in this scenario. In contrast, the KR- α method demonstrates superiority in enabling a stable RTHS with an inerter-type ES of large mass ratio, as it introduces additional numerical damping. However, it is important to note that the vRTHS with the mass ratio of 0.6 is unstable when implemented using the KR- α method where $\rho_\infty = 0.85$ (see Case 13) but becomes stable when $\rho_\infty = 0.75$ (see Case 14). It indicates that as the value of the mass ratio increases beyond 0.5 and an increasing amount of numerical damping (i.e., a smaller value of ρ_∞) is required to ensure stability.

While introducing additional numerical damping helps mitigate the spurious root-induced instability, it deteriorates the accuracy of the KR- α method, as indicated by the NRMSEs for Cases 11, 12, and 14 in Figure 15. To quantitatively assess this loss in accuracy, amplitude decay (AD) and period elongation (PE) [36] were used to characterize the numerical dispersion and energy dissipation of the KR- α method for different values of ρ_∞ when applied to RTHS with an inerter-type ES. The AD is defined as the amplitude reduction in numerically calculated displacement compared with the analytical solution for the free vibration of an undamped SDOF system:

$$m \ddot{x} + kx = 0 \quad (40)$$

with the initial condition of $x(0) = 1$ and $\dot{x}(0) = 0$. It can also be equivalently represented by the discrete equivalent damping ratio $\tilde{\zeta}$. The PE is defined as the relative difference between the exact natural period T_n ($2\pi/\omega_n$) and the discrete equivalent natural period $\tilde{T}_n$ ($2\pi/\tilde{\omega}_n$). Based on Equation (27), the $\tilde{\zeta}$ and the PE can

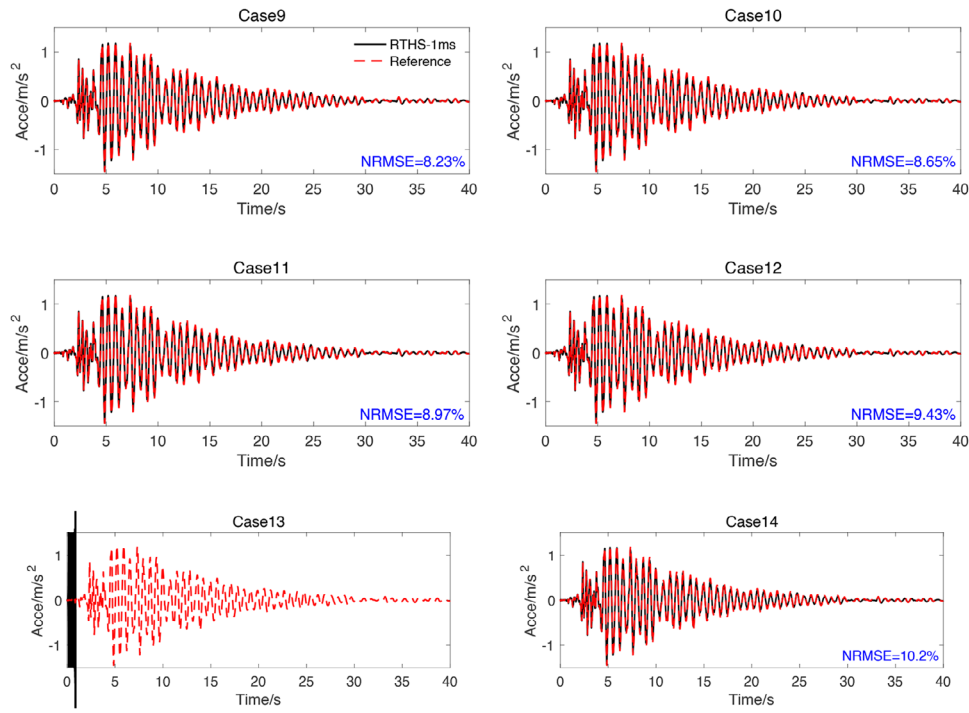


FIGURE 15 | Time histories of the acceleration in the simulations using the KR- α method.

be expressed as follows [34]:

$$\begin{aligned}\tilde{\zeta} &= \frac{-\ln(\sigma^2 + \varepsilon^2)}{2\bar{\Omega}} \\ \text{PE} &= \frac{\bar{T}_n - T_n}{T_n} = \frac{\Omega}{\bar{\Omega}} - 1 \\ \bar{\Omega} &= \bar{\omega}_n dt = \frac{\tan^{-1}(\varepsilon/\sigma)}{\sqrt{1 - \tilde{\zeta}^2}}.\end{aligned}\quad (41)$$

In Equation (41), σ and ε are the real and imaginary parts of the principal roots, respectively, obtained from the amplification matrices $\mathbf{A}$. The mass ratio of the inerter-type ES and the time delay considered in the RTHS for assessing the accuracy of the KR- α method with varying ρ_∞ is predefined as $\lambda_m = 0.5$ and $\Delta t = dt$, leading to the following discrete EOM:

$$0.5m\ddot{x}_{n+1} + kx_{n+1} + 0.5m\ddot{x}_n = 0. \quad (42)$$

In Figure 16, the AD and PE of the KR method with varying ρ_∞ are plotted against the dimensionless circular frequency Ω . The solid lines represent the errors between the numerical displacement response for the RTHS in Equation (42) and the analytical free vibration solution of Equation (40). The dotted lines depict the errors between the numerical displacement response for an ordinary time history analysis (i.e., $m\ddot{x}_{n+1} + kx_{n+1} = 0$) and the analytical free vibration solution of Equation (40). As shown in the figure, the AD and PE of the KR- α method used in RTHS with time delay are consistently larger than those for the ordinary time history analysis without time delay. Time delay is the primary cause for the integration algorithm's errors. Moreover, as the value of ρ_∞ decreases, which corresponds to an increase in numerical damping, the AD and PE of the KR- α method also

increase. This effect is more pronounced at higher dimensionless circular frequencies.

5.2 | Example 2

The above simulations for the SDOF RTHS system were then extended to the MDOF RTHS system, as depicted in Figure 17. In this case, the NS was a three-story frame, while the ESs were three inerters assumed to connect the floors and a reaction wall. The weight and lateral stiffness of each floor were defined same as those in the previous portal frame, with $W = 1000$ kN and $k = 8.22 \times 10^6$ N/m. A Rayleigh damping ratio of 2% was defined for Modes 1 and 3. The considered partitioning case in the simulation was characterized by $\lambda_m = 0.5$, $\lambda_k = 0$, and $\lambda_c = 0$ in Equation (34), with all other simulation parameters set as in the previous SDOF cases.

Figure 18 presents the acceleration time histories of the roof in simulations using the KR- α method with $\rho_\infty = 1$ and $\rho_\infty = 0.85$. It is noted that for $\rho_\infty = 1$, the KR- α method reduces to the CR method, which includes no numerical damping. As displayed, the simulation diverges when $\rho_\infty = 1$, but remains stable when $\rho_\infty = 0.85$. The critical mass ratio for the MDOF RTHS system was also found to be as 0.5, consistent with the value identified in the SDOF cases. The numerical damping is effective in mitigating the spurious root-induced instability in the conducted MDOF RTHS simulations.

5.3 | Example 3

In the third example, the neutral delayed differential equation (DDE) cases analyzed by Huang et al. [26] were simulated to fur-

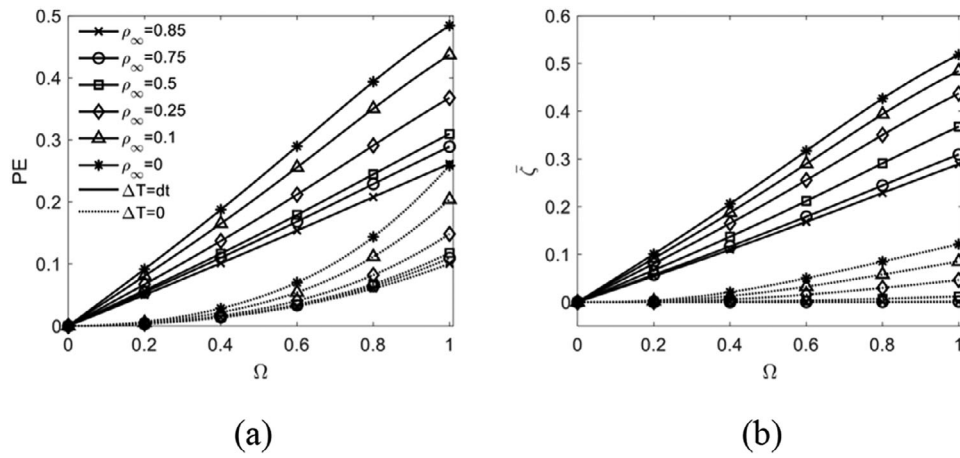


FIGURE 16 | Accuracy of the KR- α method for RTHS with an inerter-type ES of different mass ratios (a) PE (b) discrete equivalent damping ratio (i.e., the AD).

TABLE 2 | Parameters of the neutral DDE cases from ref. [26].

Type	Case #	m^N /kg	m^E /kg	Stability predicted by ref. [26]
Neutral DDE	1	0.510	0.490	Stable
	2	0.503	0.497	Stable
	3	0.501	0.499	Unstable
	4	0.500	0.500	Unstable

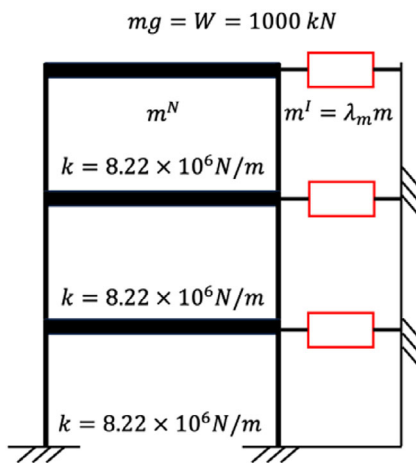


FIGURE 17 | Configuration of the MDOF RTHS.

ther substantiate the findings from the previous examples. These neutral DDE cases share similar parameters: $k^N = 157.9$ N/m, $c^N = 0.5027$ N · s/m, $k^E = 39.48$ N/m, $c^E = 0.1257$ N · s/m, and a time delay of 20 ms, with different mass partitions ($m = m^N + m^E = 1$ kg) as listed in Table 2.

In ref. [26], the stability of the neutral DDE cases was analyzed in the continuous-time domain using the Mikhailov-type stability criterion. It was concluded that for $m^E/m^N = 1$ (i.e., $\lambda_m = 0.5$), an infinite number of characteristic roots may accumulate on the imaginary axis, leading to Lyapunov stability but not asymptotic

stability. For $m^E/m^N > 1$ (i.e., $\lambda_m > 0.5$), the characteristic equation of the system must have an infinite number of roots with a positive real part, making the system unstable for any nonzero time delay. Consequently, for all roots to have only negative real parts and for no roots to accumulate on the imaginary axis, $m^E/m^N < 1$ (i.e., $\lambda_m < 0.5$) is required. These conclusions validate the critical mass ratio of $\lambda_m = 0.5$, as identified in the previous examples, for the RTHS with inerter-type ES, implemented using the CR method and the Newmark- β methods.

However, the stability analysis in ref. [26] was performed in the continuous-time domain without considering the effects of integration algorithms. As a result, the derived conclusions may not fully apply to the cases in the discrete-time domain. Based on the previous stability analysis and simulations, incorporating numerical damping helps mitigate the instability induced by spurious roots in RTHS with delayed inertial force. To further validate this claim, the vibration of neutral DDE Cases 3 and 4 in Table 1, under initial conditions $x_0 = 0$ m and $\dot{x}_0 = 1$ m/s, was simulated using the KR- α method by setting different values of ρ_∞ to introduce varying amounts of numerical damping. If the claim holds, the simulations of neutral DDE Cases 3 and 4 may remain stable even though they were assessed as unstable in ref. [26].

These simulations were conducted with a time step of 1 ms. Considering the relatively large time delay of 20 ms (i.e., $\Delta t = dt$), the values of ρ_∞ in the KR- α method were predefined as 0.5, 0.25, 0.1, and 0.05, indicating progressively higher levels of high numerical damping. The displacement responses of these simulations are presented in Figures 19 and 20. As shown, the

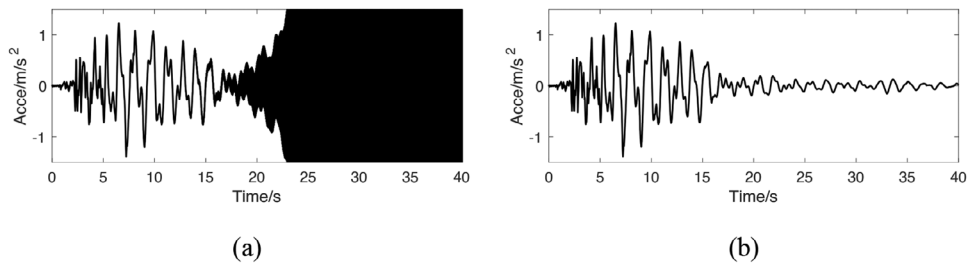


FIGURE 18 | Time histories of the roof acceleration in the simulations using the KR- α method (a) $\rho_{\infty} = 1$, (b) $\rho_{\infty} = 0.85$.

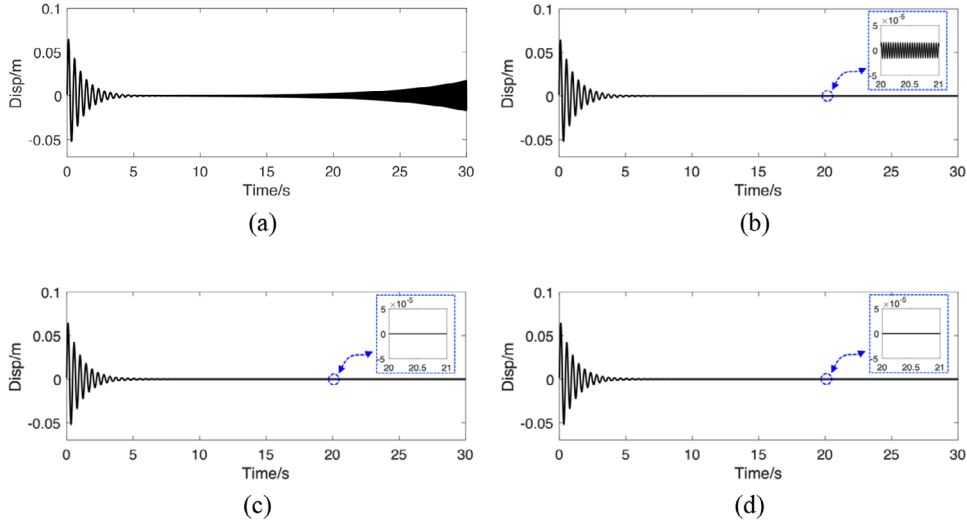


FIGURE 19 | Displacement time histories of the free vibration for neutral DDE Case 3 using the KR- α method (a) $\rho_{\infty} = 0.5$, (b) $\rho_{\infty} = 0.25$, (c) $\rho_{\infty} = 0.1$, (d) $\rho_{\infty} = 0.05$.

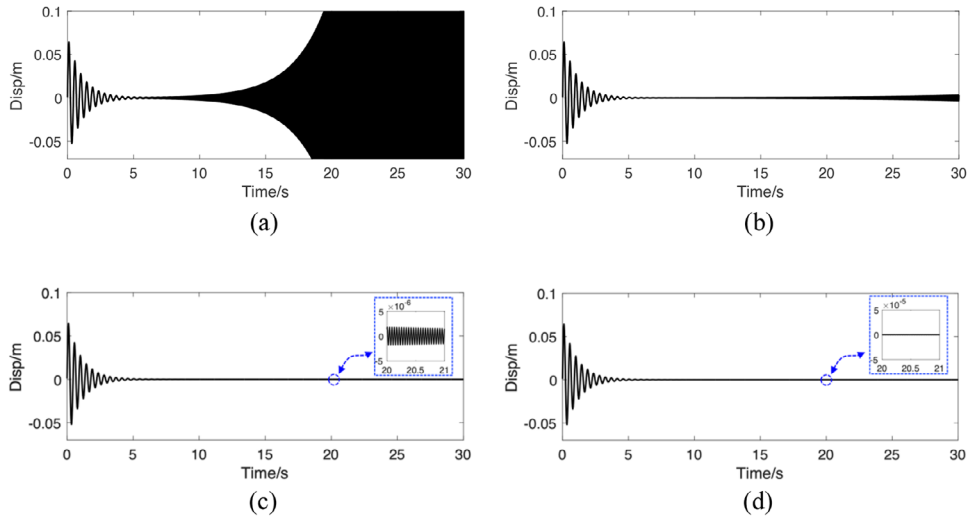


FIGURE 20 | Displacement time histories of the free vibration for neutral DDE Case 4 using the KR- α method (a) $\rho_{\infty} = 0.5$, (b) $\rho_{\infty} = 0.25$, (c) $\rho_{\infty} = 0.1$, (d) $\rho_{\infty} = 0.05$.

simulation for neutral DDE Case 3 is unstable when $\rho_{\infty} = 0.5$, and it is only marginally stable (Lyapunov stability) when $\rho_{\infty} = 0.25$, as evidenced by the oscillations in the close-up view of Figure 19b. The simulation becomes asymptotically stable only when ρ_{∞} is further reduced to 0.1 and 0.05. This trend is even more

pronounced for neutral DDE Case 4, where the system becomes asymptotically stable only when ρ_{∞} is reduced to 0.05. It is apparent from the results that incorporating numerical damping helps mitigate the instability induced by spurious roots in neutral DDE cases, where the highest-order derivative includes delay.

The neutral DDE cases previously assessed as unstable in ref. [26] in the continuous-time domain can become asymptotically stable in the discrete-time domain if sufficient numerical damping is applied. However, in the stable simulations for neutral DDE Cases 3 and 5, the required numerical damping is substantial, with ρ_∞ in the KR- α set almost to zero. This is due to the large time delay considered (20 ms), resulting in the ES responses lagging behind the NS responses by 20 time steps. In practice, the time delay can be effectively reduced to a few time steps by using a well-tuned outer-loop controller, which would consequently lower the amount of numerical damping required.

6 | Conclusions and Discussions

In this study, the stability of the RTHS, including an inerter-type ES is analyzed. The conventional understanding of the effects of time delay on RTHS stability is first reviewed using energy analysis. Subsequently, the augmented state-space equation is employed in this study to analyze the stability of RTHS considering different values of time delay. Specifically, the stability analysis is performed in the discrete-time domain for various direct integration algorithms, including the CR method, the KR- α method, and the Newmark- β method, to accurately represent the conditions of practical RTHS implementations. The effects of time delay on the stability of the discrete RTHS system with an inerter-type ES are thoroughly investigated. Additionally, a series of simulations are performed to validate the analytical investigation. The key conclusions of this paper can be summarized as follows:

1. Time delay in a discrete RTHS system implemented with direct integration algorithms results in an increased order of the system, leading to spurious roots. Specifically, in an RTHS system with an inerter-type ES, time delay decreases the magnitudes and increases the damping ratio of the principal roots. Meanwhile, it causes peaks in the magnitudes of the spurious roots, which may exceed the stability limit, making the RTHS conditionally stable. As the time delay grows, the stability region of the discrete RTHS system reduces.
2. Increasing the mass ratio (λ_m) of the ES to the NS in an RTHS leads to a consistent increase in the magnitudes of spurious roots across all frequencies, from 0 to infinite (i.e., $\Omega \in [0 \sim \infty)$). Moreover, once the mass ratio exceeds a limit, the RTHS system becomes unstable regardless of the natural frequency of the prototype structure and the time step size used for direct integration algorithms. This critical mass ratio is identified as 0.5 for the CR method and the Newmark- β method, indicating that they are unsuitable for RTHS including an inerter-type ES with $\lambda_m \geq 0.5$. However, the KR- α method can enable a stable RTHS even when $\lambda_m \geq 0.5$, by properly adjusting its numerical damping-controlled parameter ρ_∞ .
3. The effects of time delay on system stability differ between the RTHS with a stiffness-type ES and the RTHS with an inerter-type ES. The instability in the former is induced by the principal roots, whereas the instability in the latter arises from the spurious roots. While increasing inherent structural damping can mitigate the instability induced by time delay for the RTHS with a stiffness-type ES, it is ineffective for that

with an inerter-type ES. Moreover, the simulations indicate that increasing the inherent structural damping deteriorates the stability of the RTHS with an inerter-type ES. In contrast, numerical damping is found to be effective in mitigating the spurious root-induced instability in RTHS with an inerter-type ES. As the mass ratio increases, an increasing amount of numerical damping is required to ensure stability, at the cost of deteriorating the accuracy.

This study demonstrates that RTHS with an inerter-type ES can become unstable due to time delay, even though delayed inertial force is conventionally considered to dissipate energy in RTHS. The paper provides insights into using numerical damping to mitigate this instability. However, in this study, the actuator dynamics-induced tracking error in RTHS is simplified as pure time delay. The effects of amplitude errors on the performance of numerical damping in mitigating the instability in RTHS with an inerter-type ES remain unclear and require further investigation. Additionally, more realistic partitioning cases for the stability analysis of MDOF RTHS systems should be considered in future studies.

Acknowledgments

This study is based upon work funded by the State Key Laboratory of Disaster Reduction in Civil Engineering (Grant No. SLDRCE24-04) and the National Natural Science Foundation of China (Grant No. 52478124, 52378318, and 52361165658). Any opinions, findings, conclusions, and recommendations expressed here are those of the authors and do not necessarily reflect the views of the sponsors.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Appendix

The amplification matrices $\mathbf{A}$ of the discrete SDOF RTHS system using the CR method for the cases ($k > 1$):

- ($k = 2$)

$$\begin{aligned}
 \mathbf{A} &= \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{6 \times 6} \\
 \mathbf{A}_{11} &= \begin{bmatrix} 1 & dt & adt^2 \\ 0 & 1 & adt \\ -k^*/\hat{m} & -c^*/\hat{m} & -m^*/\hat{m} \end{bmatrix} \\
 \mathbf{A}_{12} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -k^E/\hat{m} & -c^E/\hat{m} & -m^E/\hat{m} \end{bmatrix}, \mathbf{A}_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\
 \mathbf{A}_{22} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned} \tag{43}$$

• ($k \geq 3$)

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \widehat{\mathbf{A}}_{11} & \mathbf{A}_{12} \\ \widehat{\mathbf{A}}_{21} & \widehat{\mathbf{A}}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{3k \times 3k} \\ \widehat{\mathbf{A}}_{11} &= [\mathbf{A}_{11} \quad 0], \widehat{\mathbf{A}}_{11} \in \mathbb{R}^{3 \times 3(k-1)} \\ \widehat{\mathbf{A}}_{21} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}, \widehat{\mathbf{A}}_{21} \in \mathbb{R}^{3(k-1) \times 3(k-1)}, \\ \widehat{\mathbf{A}}_{22} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 0 \end{bmatrix}, \widehat{\mathbf{A}}_{22} \in \mathbb{R}^{3(k-1) \times 3} \end{aligned} \quad (44)$$

The amplification matrices $\mathbf{A}$ of the discrete SDOF RTHS system using the KR- α method for the cases ($k > 1$):

• ($k = 2$)

• ($k = 2$)

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{6 \times 6} \\ \mathbf{A}_{11} &= \begin{bmatrix} 1 - (\beta k^* dt^2) / \widehat{m} & dt - (\beta c^* dt^2) / \widehat{m} & (0.5 - \beta) dt^2 - (\beta m^* dt^2) / \widehat{m} \\ -(\gamma k^* dt) / \widehat{m} & 1 - (\gamma c^* dt) / \widehat{m} & \gamma dt - \gamma m^* dt / \widehat{m} \\ -k^* / \widehat{m} & -c^* / \widehat{m} & -m^* / \widehat{m} \end{bmatrix} \\ \mathbf{A}_{12} &= \begin{bmatrix} -(\beta k^E dt^2) / \widehat{m} & -(\beta c^E dt^2) / \widehat{m} & -(\beta m^E dt^2) / \widehat{m} \\ -(\gamma k^E dt) / \widehat{m} & -(\gamma c^E dt) / \widehat{m} & -\gamma m^E dt / \widehat{m} \\ -k^E / \widehat{m} & -c^E / \widehat{m} & -m^E / \widehat{m} \end{bmatrix} \\ \mathbf{A}_{21} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{A}_{22} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned} \quad (47)$$

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \widehat{\mathbf{A}}_{11} & \mathbf{A}_{12} \\ \widehat{\mathbf{A}}_{21} & \widehat{\mathbf{A}}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{9 \times 9} \\ \widehat{\mathbf{A}}_{11} &= \begin{bmatrix} \widehat{\mathbf{A}}_{11}^1 & \widehat{\mathbf{A}}_{11}^2 \end{bmatrix}, \widehat{\mathbf{A}}_{11} \in \mathbb{R}^{3 \times 6} \\ \widehat{\mathbf{A}}_{11}^1 &= \begin{bmatrix} 1 & dt & \alpha_2 dt^2 \\ 0 & 1 & \alpha_1 dt \\ \frac{k^N}{m^N (\alpha_3 - 1)} & \frac{c^N - k^N (\alpha_f - 1) dt}{m^N (\alpha_3 - 1)} & \frac{m^N \alpha_3 - k^N (\alpha_f - 1) \alpha_2 dt^2 - c^N (\alpha_f - 1) \alpha_1 dt}{m^N (\alpha_3 - 1)} \end{bmatrix} \\ \widehat{\mathbf{A}}_{11}^2 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{-k^E (\alpha_f - 1)}{m^N (\alpha_3 - 1)} & \frac{-c^E (\alpha_f - 1)}{m^N (\alpha_3 - 1)} & \frac{-m^E (\alpha_f - 1)}{m^N (\alpha_3 - 1)} \end{bmatrix} \\ \mathbf{A}_{12} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{k^E \alpha_f}{m^N (\alpha_3 - 1)} & \frac{c^E \alpha_f}{m^N (\alpha_3 - 1)} & \frac{m^E \alpha_f}{m^N (\alpha_3 - 1)} \end{bmatrix} \\ \widehat{\mathbf{A}}_{21} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}, \widehat{\mathbf{A}}_{21} \in \mathbb{R}^{6 \times 6}, \widehat{\mathbf{A}}_{22} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 0 \end{bmatrix}, \widehat{\mathbf{A}}_{22} \in \mathbb{R}^{6 \times 3} \end{aligned} \quad (45)$$

• ($k \geq 3$)

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \widehat{\mathbf{A}}_{11}^* & \mathbf{A}_{12} \\ \widehat{\mathbf{A}}_{21}^* & \widehat{\mathbf{A}}_{22}^* \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{3(k+1) \times 3(k+1)} \\ \widehat{\mathbf{A}}_{11}^* &= \begin{bmatrix} \widehat{\mathbf{A}}_{11}^1 & 0 & \widehat{\mathbf{A}}_{11}^2 \end{bmatrix}, \widehat{\mathbf{A}}_{11}^* \in \mathbb{R}^{3 \times 3k} \\ \widehat{\mathbf{A}}_{21}^* &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}, \widehat{\mathbf{A}}_{21}^* \in \mathbb{R}^{3k \times 3k}, \widehat{\mathbf{A}}_{22} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 0 \end{bmatrix}, \widehat{\mathbf{A}}_{22} \in \mathbb{R}^{3k \times 3} \end{aligned} \quad (46)$$

The amplification matrices $\mathbf{A}$ of the discrete SDOF RTHS system using the Newmark- β method for the cases ($k > 1$):

• ($k \geq 3$)

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \widehat{\mathbf{A}}_{11} & \mathbf{A}_{12} \\ \widehat{\mathbf{A}}_{21} & \widehat{\mathbf{A}}_{22} \end{bmatrix}, \mathbf{A} \in \mathbb{R}^{3k \times 3k} \\ \widehat{\mathbf{A}}_{11} &= [\mathbf{A}_{11} \quad 0], \widehat{\mathbf{A}}_{11} \in \mathbb{R}^{3 \times 3(k-1)} \\ \widehat{\mathbf{A}}_{21} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}, \widehat{\mathbf{A}}_{21} \in \mathbb{R}^{3(k-1) \times 3(k-1)}, \\ \widehat{\mathbf{A}}_{22} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 0 \end{bmatrix}, \widehat{\mathbf{A}}_{22} \in \mathbb{R}^{3(k-1) \times 3} \end{aligned} \quad (48)$$